

Candidate Local GR Completion from the Existing Complex-Time Profile

Free-embedding lift, pure- Θ Gauss action, and exact vacuum closure

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Abstract

The torsion-free single-section construction $E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta \in W_L$ obeys a concurrent-vector no-go and cannot generate a generic non-flat vacuum geometry. This note proves a precise alternative using no new fundamental field: retain the already postulated periodic complex-time profile $\psi \mapsto \Theta(x, \psi)$ and read the four-dimensional metric from the normalized Haar average of the central Lorentz product. A local free isometric embedding of any Lorentzian four-metric into $\mathbb{E}^{13,1}$ lifts isometrically to fourteen orthonormal W_L -valued profile modes of the same Θ . The ten normal second-derivative profiles are then independent, so variation of a Gauss functional algebraically identical to the composite Einstein–Hilbert– Λ action is exactly equivalent to the vacuum Einstein– Λ equations on that branch. This closes local representability and establishes conditional local vacuum equivalence for the adopted profile metric and action. It does not derive κ , Λ , profile selection or the profile readout from the previous canonical master action. A companion note supplies a flat ambient profile connection whose tangential projection is the induced Levi–Civita connection, but identifying that ambient derivative with the presently overloaded canonical symbol D_μ still requires a deliberate architectural revision rather than a silent status change.

Canonical impact

The construction below uses one profile-valued field and adds no independent metric, connection, or tetrad. It is nevertheless *not compatible with the current wording of AXIOM C*, which forbids compact- ψ averaging as the canonical metric readout. The fixed-frame formulas can be made covariant by placing the profiles in a flat pairing-compatible ambient bundle and replacing ∂_μ by its flat ambient connection $\mathbb{D}_\mu$; the induced tangential connection is then Levi–Civita. This is proved in `gap_10s_covariant_profile_geometry.tex`. It does not, however, coincide with the currently canonical pointwise spin-lift interpretation of D_μ . The result is therefore a noncanonical conditional completion candidate, not yet a replacement axiom.

1 Central profile pairing

Let

$$W_L = \{ix^0 \mathbf{1} + x^k \mathbf{e}_k \mid x^a \in \mathbb{R}\} \subset \mathbb{B}$$

be the canonical Lorentz slice. For $U, V : S_{R_\psi}^1 \rightarrow W_L$ define the real symmetric profile pairing by the central identity

$$\langle U, V \rangle_\psi \mathbf{1} := \frac{1}{L_\psi} \int_0^{L_\psi} \frac{1}{2} (U^\# V + V^\# U) \, d\psi, \quad L_\psi = 2\pi R_\psi. \quad (1)$$

It is the orthogonal direct sum of countably many copies of the Lorentz form on W_L and therefore has infinitely many positive and negative directions.

Definition 1.1 (Candidate profile metric). *For a smooth periodic field $\Theta : U \times S_{R_\psi}^1 \rightarrow W_L$ set*

$$E_\mu := \partial_\mu \Theta, \quad \boxed{g_{\mu\nu}[\Theta] = \frac{1}{\mathcal{N}_0} \langle E_\mu, E_\nu \rangle_\psi.} \quad (2)$$

Equivalently, $g_{\mu\nu} \mathbf{1}$ is the normalized Haar average of the pointwise central anticommutator. No trace or real-part projection is used.

Proposition 1.2 (Finite Lorentz ambient block inside one profile). *There exist fourteen smooth periodic profiles $F_A : S^1 \rightarrow W_L$, $A = 0, \dots, 13$, satisfying*

$$\langle F_A, F_B \rangle_\psi = \eta_{AB}, \quad \eta_{AB} = \text{diag}(-1, +1, \dots, +1). \quad (3)$$

Proof. Take a unit timelike constant profile for F_0 . For the positive directions use normalized sine/cosine profiles in the three spacelike quaternion basis directions, with distinct pairs (Fourier number, algebra direction). Fourier orthogonality makes different choices orthogonal, while the Lorentz slice identity fixes every norm to +1. There are arbitrarily many such positive profiles, so thirteen can be selected. $\square$

Let $\iota : \mathbb{R}^{13,1} \rightarrow \mathcal{V}_\psi$ denote the isometric linear map

$$\iota(v) := v^A F_A, \quad \langle \iota(v), \iota(w) \rangle_\psi = \eta_{AB} v^A w^B.$$

2 Free-embedding lift

Definition 2.1 (Free isometric embedding). *A local isometric embedding $Y : U \rightarrow \mathbb{E}^{13,1}$ is free when the fourteen ambient vectors*

$$\{Y_{,\mu}, Y_{;\mu\nu}\}, \quad \mu = 0, \dots, 3, \quad \mu \leq \nu, \quad (4)$$

are linearly independent. The ten $Y_{;\mu\nu}$ are normal to the four $Y_{,\mu}$.

Theorem 2.2 (Profile lift of a free embedding). *Let $Y : U \rightarrow \mathbb{E}^{13,1}$ be a local free isometric embedding of a Lorentzian metric g . Define one biquaternionic profile field*

$$\boxed{\Theta_Y(x, \psi) := \sqrt{\mathcal{N}_0} Y^A(x) F_A(\psi).} \quad (5)$$

Then:

1. $g_{\mu\nu}[\Theta_Y] = g_{\mu\nu}$ exactly;
2. the normal profile second derivatives are

$$B_{\mu\nu} := P_\perp(\nabla_\mu E_\nu) = \sqrt{\mathcal{N}_0} \iota(Y_{;\mu\nu}); \quad (6)$$

3. the closure map

$$\mathcal{K}(X) = X^{\mu\nu} B_{\mu\nu} \quad (7)$$

is injective.

Proof. Differentiating Eq. (5) gives $E_\mu = \sqrt{\mathcal{N}_0} \iota(Y_{,\mu})$. The isometry of ι yields

$$\mathcal{N}_0^{-1} \langle E_\mu, E_\nu \rangle_\psi = \eta_{AB} Y^A_{,\mu} Y^B_{,\nu} = g_{\mu\nu}.$$

Because ι is constant and isometric, it commutes with spacetime covariant differentiation and with orthogonal decomposition relative to the tangent span. Hence Eq. (6) holds. Freeness makes the ten normal vectors $Y_{;\mu\nu}$ linearly independent, and the injectivity of ι preserves that independence. Therefore $\mathcal{K}(X) = 0$ implies $X^{\mu\nu} = 0$. $\square$

Theorem 2.3 (Local representability). *Every four-dimensional Lorentzian manifold of the regularity covered by the local free-embedding theorem admits, near each point, a field of the form Eq. (5) that reproduces its metric and is fiber-free.*

Proof. For a four-dimensional Lorentzian metric of class C^m with $m \geq 3$, or C^∞ , Greene’s local free-isometric-embedding theorem (as summarized in Ref. [3]) gives a free isometric embedding $Y : U \rightarrow \mathbb{E}^{13,1}$. Apply the profile-lift theorem. $\square$

Corollary 2.4 (Local Schwarzschild and general Einstein patches). *Every regular coordinate patch of Schwarzschild, Kerr, FLRW, or any other smooth Lorentzian Einstein solution has a local fiber-free Θ_Y representer in the candidate profile metric. This is an existence theorem; it does not supply a preferred or global closed-form representer.*

3 Gauss identity and a direct pure- Θ action

The target profile space has a fixed flat pairing. Its Gauss equation gives

$$R_{\mu\nu\rho\sigma} = \frac{1}{\mathcal{N}_0} \left(\langle B_{\mu\rho}, B_{\nu\sigma} \rangle_\psi - \langle B_{\mu\sigma}, B_{\nu\rho} \rangle_\psi \right). \quad (8)$$

Consequently

$$R[g[\Theta]] = \frac{1}{\mathcal{N}_0} \left(\langle B^\mu{}_\mu, B^\nu{}_\nu \rangle_\psi - \langle B_{\mu\nu}, B^{\mu\nu} \rangle_\psi \right). \quad (9)$$

Definition 3.1 (Pure- Θ Gauss action). *On the regular profile branch define*

$$S_G[\Theta] = \frac{1}{2\kappa} \int_U \sqrt{-g[\Theta]} \left[\frac{1}{\mathcal{N}_0} \left(\langle B^\mu{}_\mu, B^\nu{}_\nu \rangle_\psi - \langle B_{\mu\nu}, B^{\mu\nu} \rangle_\psi \right) - 2\Lambda \right] d^4x. \quad (10)$$

The usual composite boundary term is understood, or variations are taken with compact support.

Proposition 3.2 (No independent metric or connection). *On every regular profile configuration, Eq. (10) equals the Einstein–Hilbert– Λ functional of the composite metric $g[\Theta]$. All objects in Eq. (10) are built from one Θ , the fixed profile pairing, and spacetime derivatives.*

Theorem 3.3 (Exact local vacuum GR closure). *Let Θ be a regular fiber-free stationary point of $S_G[\Theta]$. Then*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 0. \quad (11)$$

Conversely, every local free profile lift of an Einstein– Λ metric is a stationary point under compactly supported profile variations.

Proof. The metric variation is

$$\delta g_{\mu\nu} = \frac{2}{\mathcal{N}_0} \langle E_{(\mu}, \partial_{\nu)} \delta \Theta \rangle_\psi.$$

Varying the composite Einstein–Hilbert functional, integrating by parts, and using the contracted Bianchi identity gives the normal Euler–Lagrange equation

$$(G^{\mu\nu} + \Lambda g^{\mu\nu}) B_{\mu\nu} = 0. \quad (12)$$

Fiber-freeness makes the ten $B_{\mu\nu}$ independent, so Eq. (12) implies Eq. (11). The converse follows immediately because the Einstein residual vanishes before contraction with $B_{\mu\nu}$. $\square$

4 Compatibility conditions before canonical promotion

The closure theorem above is an exact statement inside the adopted embedding architecture, but three compatibility questions must be separated from it.

Covariant ambient jet. The companion note `gap_10s_covariant_profile_geometry.tex` places the profiles in a flat pairing-compatible ambient bundle with connection $\mathbb{D}_\mu$. In a local parallel frame $\mathbb{D}_\mu = \partial_\mu$, while in arbitrary profile frames the metric, normal bundle and free rank are gauge invariant. The tangential projection of $\mathbb{D}_\mu$ is exactly the induced Levi–Civita connection, and the pointwise concurrent-vector no-go does not apply. What remains open is canonical symbol and action reconciliation: the old pointwise spin-lift derivative D_μ must be distinguished from the flat ambient derivative $\mathbb{D}_\mu$, not silently identified with it.

Fourteen-mode sector. The full profile space contains more directions, including additional timelike Fourier directions. The proof embeds a finite $\mathbb{E}^{13,1}$ block inside that space. A canonical theory must either derive a stable fourteen-mode sector, prove that the remaining directions are pure gauge/auxiliary, or perform a constraint analysis showing that they add no physical ghost or bending modes.

Action origin. The Gauss functional in Eq. (10) is the Einstein–Hilbert– Λ functional rewritten in profile variables. Its variation proves equivalence after this action is adopted; this is not yet a derivation from the older UBT master functional. Deriving that action is the remaining dynamical bridge.

5 What is closed and what remains

Item	Status	Scope
Local metric representability	Closed (profile metric)	Follows from local free embedding in $\mathbb{E}^{13,1}$ and the profile lift.
Fiber-free non-vacuity on GR geometries	Closed locally	Every locally free embedded GR patch lifts to one Θ .
Vacuum Einstein- Λ equation	Conditionally closed (free branch)	Exact after adopting the profile metric and the composite Gauss/EH action.
Independent tetrad/connection fields	None introduced in this branch	g , B , and the Levi-Civita objects are composites of Θ .
Covariant profile-frame realization	Conditionally closed	A flat pairing-compatible ambient connection preserves the metric and free rank; canonical identification with the old pointwise D_μ remains open.
Fourteen-mode selection / extra profiles	Open	Must derive a stable finite block or prove the remaining profile directions nonphysical.
Canonical metric choice	Axiom and compatibility decision	Current AXIOM C explicitly forbids compact- ψ averaging.
Origin of κ and Λ	Open	The Gauss action fixes the form, not their values from the older master equation.
Dynamic profile selection/stability	Open	Must show the UBT dynamics selects and preserves the free, ψ -stable branch.
Holomorphic/Jacobi restriction	Open	The lift is smooth periodic; any stronger analytic restriction must be checked separately.
Global representability	Open	The theorem is local; topology and horizons require patching/global analysis.
Matter and gauge separation	Open	Vacuum needs no separate $\mathcal{F}_\Theta$ term; unified matter requires a projection/separation theorem.

6 Decision

The candidate profile completion is not merely a right inverse for a prescribed tetrad. Once the profile metric and the Gauss action are adopted, the field Θ is the embedding variable, its stationary equation is Eq. (12), and freeness converts that equation to the complete ten Einstein equations. The construction therefore supplies an exact local vacuum GR sector from one Θ .

The rank and local embedding calculation are no longer the principal obstacle. The remaining decision is whether the profile architecture can be derived as a genuine UBT sector: its scalar-profile derivative must be distinguished from the canonical pointwise spin lift, the extra profile directions must be controlled, and the Gauss/EH functional must follow from the master dynamics. Only after those bridges are supplied should AXIOM C be replaced. The pointwise torsion-free route has the concurrent-vector no-go; the profile route has exact conditional local equivalence, not yet an unconditional master-action derivation.

References

- [1] M. D. Bustamante, P. Debbasch, and M. E. Brachet, “Classical Gravitation as Free Membrane Dynamics,” *Class. Quantum Grav.* 23 (2006), arXiv:gr-qc/0509090.
- [2] R. E. Greene, *Isometric Embeddings of Riemannian and Pseudo-Riemannian Manifolds*, Memoirs of the American Mathematical Society, No. 97 (1970).
- [3] S. Willison, “A Re-examination of the Isometric Embedding Approach to General Relativity,” arXiv:1311.6203.